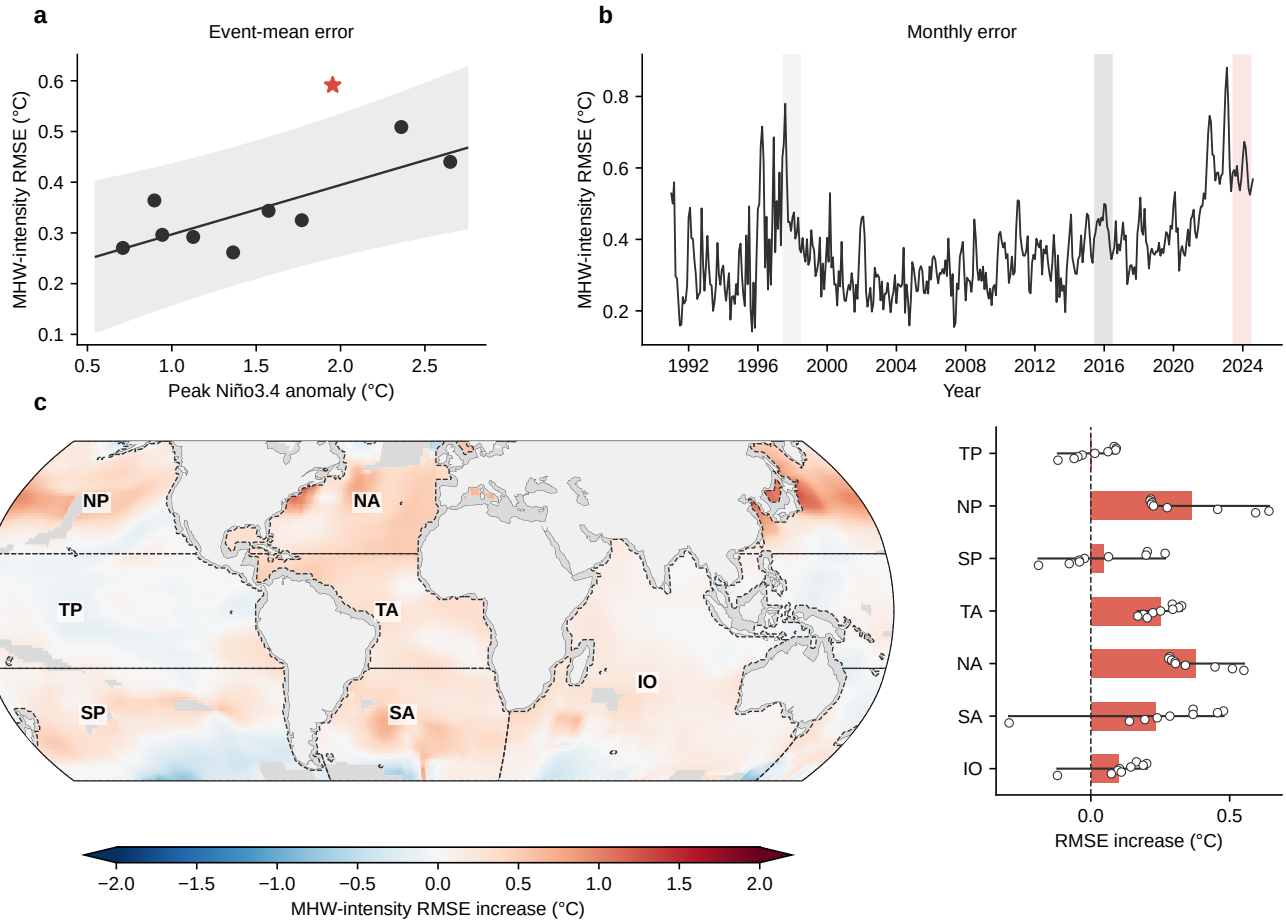
Supplementary Fig. 1 | MHW-intensity forecast errors in time and space



a, Event-mean MHW-intensity root-mean-square error (RMSE) versus peak Niño3.4 anomaly. The regression and grey 95% prediction interval use nine earlier El Niño events; the red star marks 2023–2024, excluded from fitting. b, Monthly global intensity RMSE (404 months); shading marks the three strong events. c, Intensity RMSE in 2023–2024 minus the reference from 1997–1998 and 2015–2016, using 18-month windows and 11° × 11° same-basin neighbourhoods. Red indicates larger errors and blue smaller errors; grey ocean cells lack a valid comparison. The inset summarizes all seven analysis regions, retaining all nine leads and their range. TP, tropical Pacific; NP, North Pacific; SP, South Pacific; TA, tropical Atlantic; NA, North Atlantic; SA, South Atlantic; IO, Indian Ocean. Intensity is positive SST excess above the local 1985–2014 calendar-month 90th percentile. Verification includes only observed-MHW samples: smaller RMSE indicates better intensity prediction but does not assess false alarms outside observed MHWs.

Spatial windows run from April of the event year to September of the next year. The reference RMSE is the root of the equal-event mean squared error; regional summaries also aggregate squared errors before taking roots. Lead ranges are not confidence intervals. Panels a,b retain monthly spatial aggregation. The 1° display grid does not imply independent 1° information. Coverage and pooling sensitivity are shown in Supplementary Figs. 16–17.